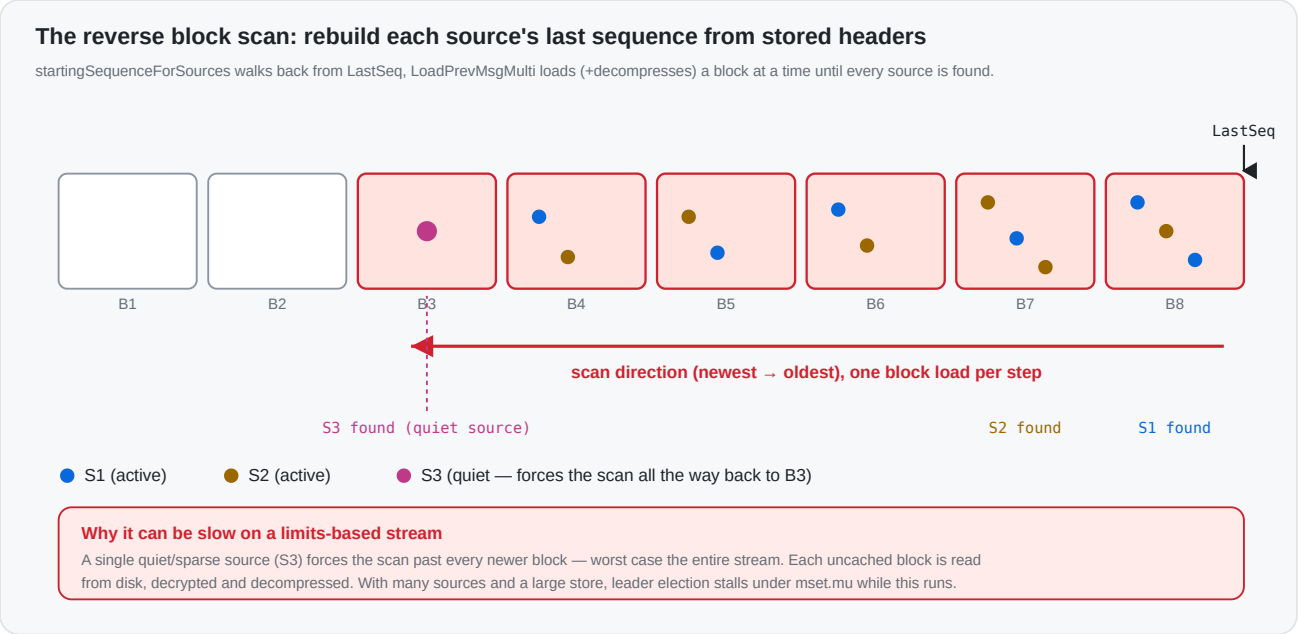
Proposal: Replace the sourcing reverse block scan with an index lookup

Status	Proposal
Area	JetStream stream sourcing (`server/stream.go`) + store index (`server/filestore.go` , `server/memstore.go`)
Date	2026-06-06
Companion	`sourcing-durable-resume.md` (lifecycle, durable consumer, triggers)
Target	`startingSequenceForSources` (`stream.go:4694`) and `setStartingSequenceForSources` (`stream.go:4566`)

1. The problem in one paragraph

On hub-stream **leader election / restart**, `setupSourceConsumers` calls `startingSequenceForSources` **unconditionally** (`stream.go:4821`). That function reconstructs each source's last sourced sequence by walking the stream's **own store backwards from** `LastSeq` , one block at a time (`LoadPrevMsgMulti` , `filestore.go:9425`), reading the `Nats-Stream-Source` header off matching messages, until **every** source has been located.

> Note: the walk is not as naive as message-by-message — within each block the per-block index (`fss`) skips non-matching messages, and the per-source sublist narrows as sources are found. But `LoadPrevMsgMulti` still **visits every block** from `LastSeq` back to the match (loading/decompressing cold ones), and the outer loop **re-walks** as the sublist narrows. So the cost grows with both the number of blocks back to the least-recently-active source (worst case the **entire store**) and the number of sources — all under `mset.mu` .



2. The key insight

The store **already maintains a per-subject index** that can return the last sequence for a subject *without* walking messages:

- `fs.psim` — per-subject info, including `lblk` = the **last block index** that contains each subject (`filestore.go:3843-3852` , `8980-8984`).
- `mb.fss` — per-block subject state with `Last` per subject (`filestore.go:9003-9015`).

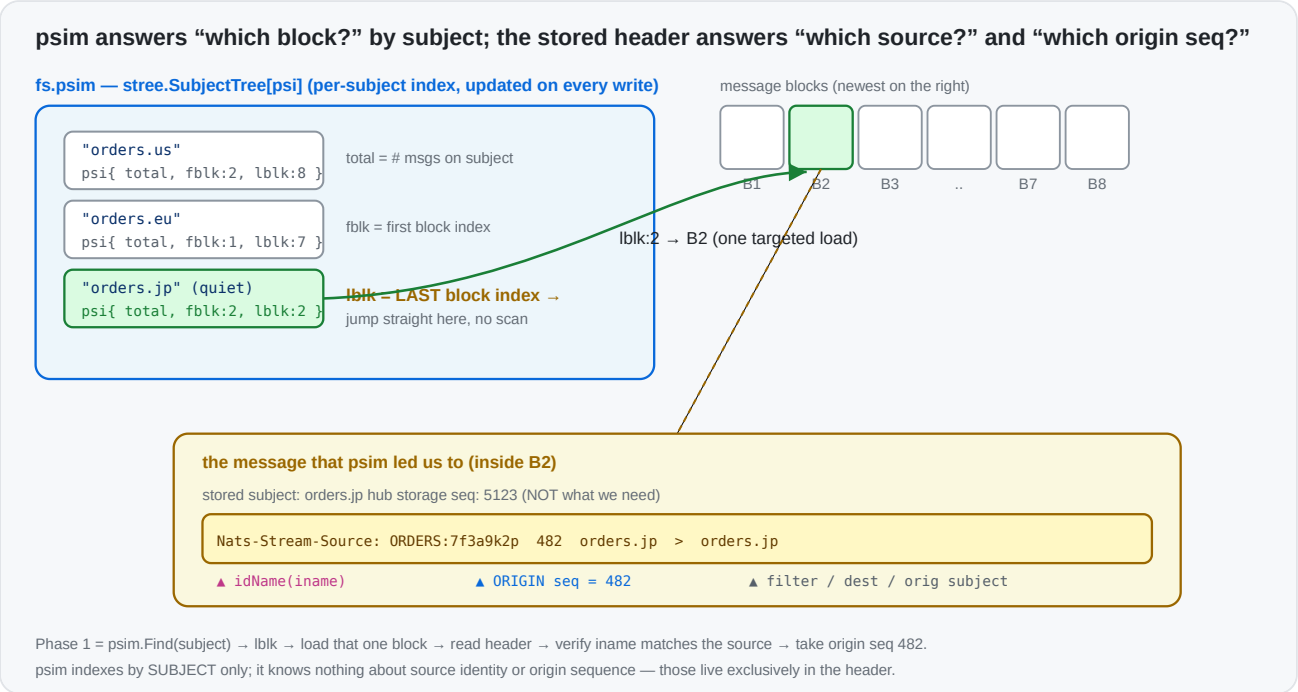
This is what powers `LoadLastMsg(subject)` → `loadLast` (`filestore.go:8939`): for a literal subject it jumps straight to `info.lblk` and reads that subject's last message — **one targeted block load**, regardless of how far back it is. `MultiLastSeqs` (`filestore.go:3806`) does the same for a batch of filters in a single index pass.

The current scan ignores this index and instead does a generic backward message walk. The header it needs (the **origin** sequence) is right there in the message that `LoadLastMsg` already returns.

> Why a plain "last seq for subject" isn't enough on its own: `si.sseq` must be the **origin** stream sequence taken from the `Nats-Stream-Source` header, not the hub's storage sequence. So we use the index to *find* the source's last stored message in O(1), then read the origin seq from its header.

3. Background: the subject index (`psim` + `fss`) and the source header

The filestore keeps a **two-level subject index**. Phase 1 stands entirely on it, so it's worth being precise about what each level is.



3.1 `psim` — the store-wide "which blocks hold this subject" map

`fs.psim` (`filestore.go:195`) is a `stree.SubjectTree[psi]` — a subject-token tree keyed by the full message subject — whose value is a tiny per-subject record (`filestore.go:168`):

```
type psi struct {
    total uint64  // how many messages exist for this subject across the whole s
    fblk  uint32  // index of the FIRST block that still contains this subject
    lblk  uint32  // index of the LAST  block that still contains this subject
}
```

It is maintained incrementally on every write (`filestore.go:4933-4943`): on store, the subject's entry is created (`total:1, fblk=lblk=current block`) or its `total` is bumped and `lblk` advanced to the current block; on removal/expiry the counts and `fblk` / `lblk` are walked forward as blocks empty out. Because `psim` is a *subject tree*, it supports both exact `Find(subject)` and wildcard `Match(filter)` in time proportional to the subject token structure — **not** to the number of messages.

The single field that makes Phase 1 cheap is `lblk`: given a source's subject, `psim.Find` returns the **exact last block** that contains it. There is no need to scan newer blocks — most of which don't contain that subject at all. (`MultiLastSeqs`, `filestore.go:3806`, uses the same `psim` to answer a *batch* of filters in one index pass.)

3.2 `fss` — the per-block "first/last seq of this subject in this block" map

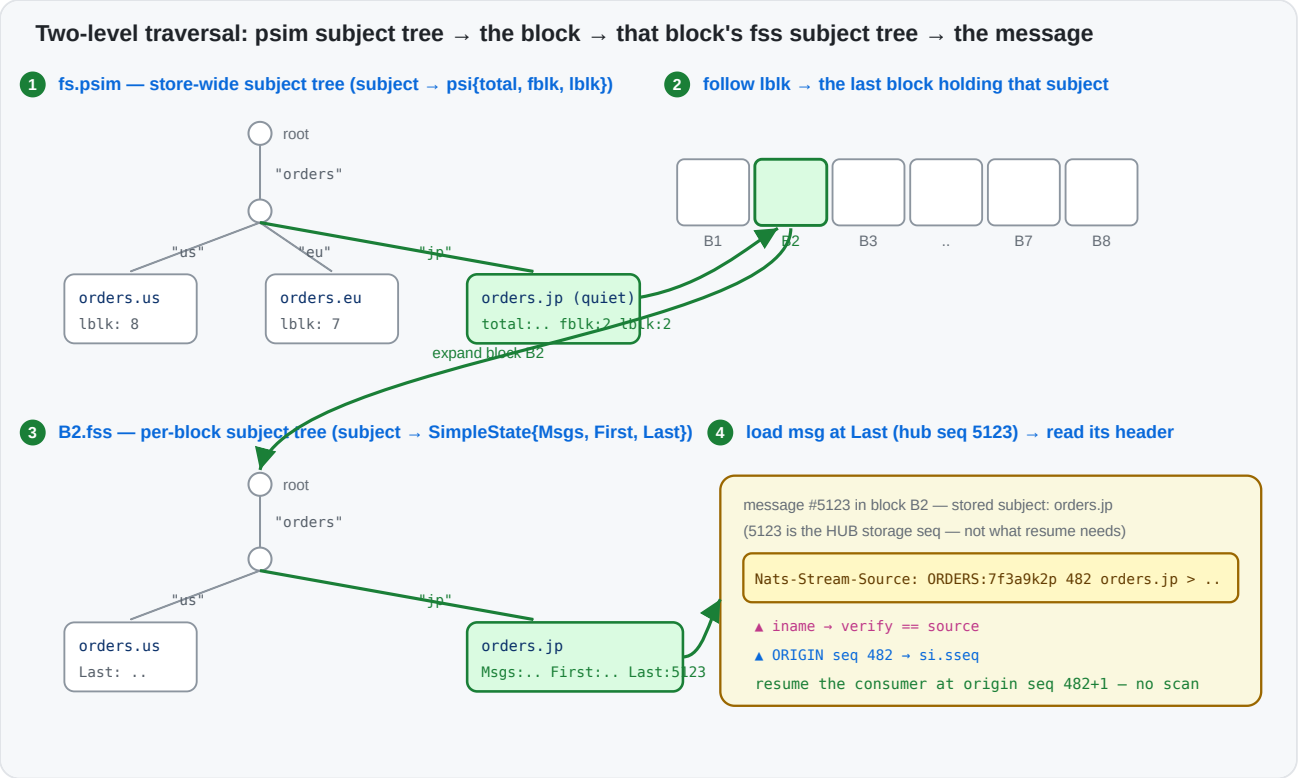
`psim` gets us to a block; `mb.fss` (`filestore.go:239`) finishes the job *inside* that block. Each `msgBlock` carries its own `stree.SubjectTree[SimpleState]` mapping subject → (`store.go:180`):

```
type SimpleState struct {
    Msgs  uint64 // messages for this subject in THIS block
    First uint64 // first sequence for this subject in this block
    Last  uint64 // last  sequence for this subject in this block
    // (firstNeedsUpdate / lastNeedsUpdate: lazily recomputed when interior dele
}
```

So the two levels compose into an O(1)-ish lookup, which is exactly what `loadLast` (`filestore.go:8980-9034`) does:

```
psim.Find(subject) ─► info.lblk ─► block ─► block.fss.Find(subject) ─► ss.Last -
```

`fss` is what gives us the precise sequence within the block (and it correctly skips interior-deleted messages via `lastNeedsUpdate` / `recalculateForSubj`). On a cold block, `ensurePerSubjectInfoLoaded` loads/derives `fss` for that **one** block — the cost we pay once per resolved source, versus the current scan paying it for every block back to the oldest source.



3.3 Why each message carries a `Nats-Stream-Source` header

`psim` / `fss` index by **subject**. They know nothing about *which source* produced a stored message or *where in the origin stream* it came from — and those are precisely the two facts resume needs. That information is carried only in the per-message header `Nats-Stream-Source` (`JSStreamSource` , `stream.go:635`), written by `genSourceHeader` (`stream.go:4470`) as:

```
Nats-Stream-Source:  <idName>  <originSeq>  <filter>  <destination>  <origSubjec
                       |          |
                       |          └─ the ORIGIN stream's sequence — what we must
                       └─ origin stream name (+ domain/consumer hash); with filt
                          reconstructs the source's full iname  (parsed by strea
```

This header is needed for three independent reasons:

1. **Source attribution.** A hub message is stored under its *destination* subject; the storage layer records nothing about its origin. When two sources can land on the same/overlapping subject, only the header's `iname` says which source a given message belongs to — hence Phase 1's `iname == si.iname` check.
2. **Origin-sequence tracking.** The sourcing consumer is (re)created on the **origin** with `DeliverByStartSequence`, which lives in the *origin's* sequence space. The hub's own storage sequence (e.g. 5123) is unrelated to the origin sequence (e.g. 482). Only the

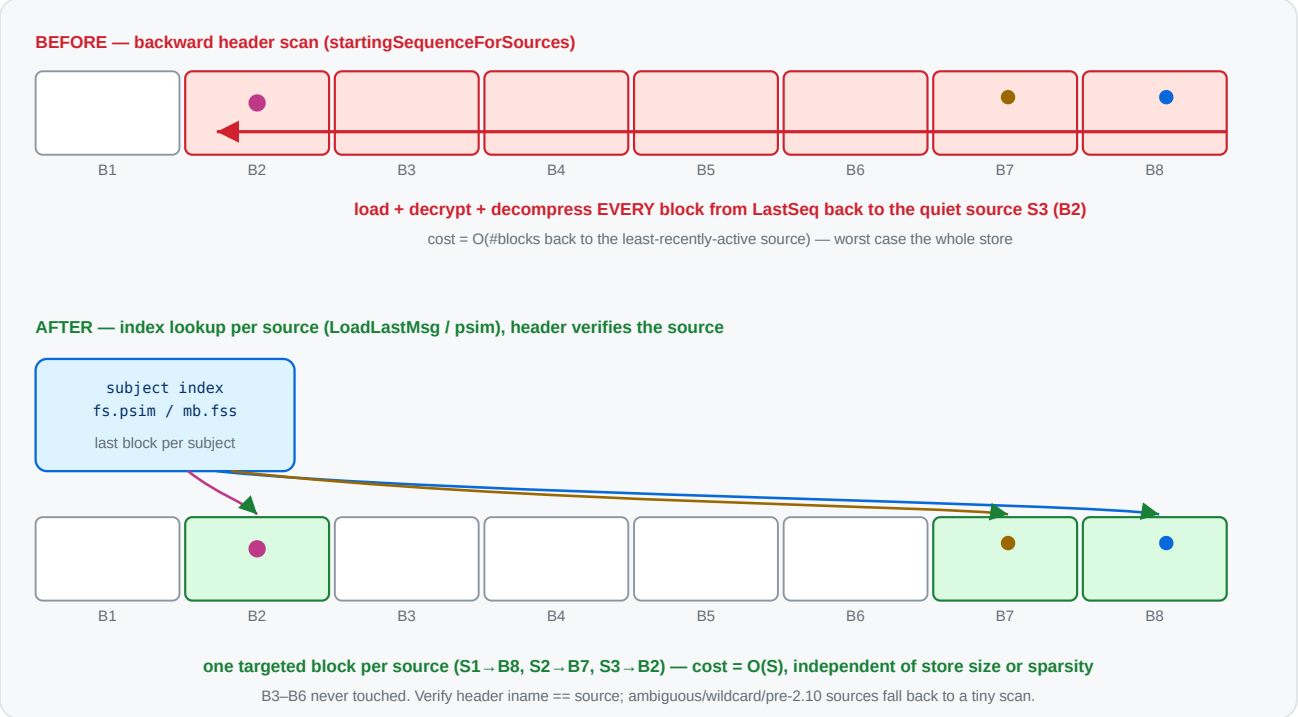
header preserves the origin sequence, so it is the *only* place a resume point can be recovered from without contacting the origin.

3. **Loop / daisy-chain handling & dedup.** When re-sourcing (A→B→C), `processInboundSourceMsg` strips the inbound `JSStreamSource` and stamps its own (`stream.go:4398`, `4405`), so each hop's provenance is well-defined and re-sourced duplicates can be recognised.

This is the crux of the whole proposal: **the index tells us *which block* in O(1); the header (read from that one message) tells us *which source* and *which origin seq*.** Today's scan reads the header off *every* message on the way back because it never consults `psim` / `fss` to jump directly to the right one.

4. Proposed algorithm

Two phases. Phase 1 resolves the common case with index lookups; Phase 2 keeps today's scan only for sources that genuinely can't be attributed by subject.



For each source, its **stored subject(s)** are: the `FilterSubject` for a plain source, or the transform **Destination(s)** for a transformed source (this mirrors how the current sublist is built at `stream.go:4742-4757`).

```
Phase 1 — index lookup (per source, O(1) targeted load):
  for each source si with concrete stored subject sf (not "" / not ">"):
      sm := store.LoadLastMsg(sf)                    // jumps via psim, ~1 block
      if sm has a JSStreamSource header:
```

```
          (_, iname, sseq) := streamAndSeq(header)
          if iname == si.iname:                    // self-verifying attribution
              si.sseq = sseq; mark resolved
  // anything not resolved (no header / header for a different source / wildcard

Phase 2 — fallback reverse scan (only for the unresolved subset):
  run today's LoadPrevMsgMulti loop, but with the sublist built from ONLY the
  unresolved sources, so it narrows immediately and stops as soon as they are fo
```

The header check makes Phase 1 **self-correcting**: if a subject is shared between sources, or overlaps a direct publish on the hub, or the last message has no source header, the lookup simply doesn't match and that source drops to Phase 2 — never producing a wrong answer.

5. Before → after (code)

Before (`startingSequenceForSources` , condensed — `stream.go:4694`)

```
mset.resetSourceInfo()
// build a sublist of ALL sources' subjects
refreshSublist()                                  // every source
for last := state.LastSeq; ; {                    // walk the WHOLE store backw
    sm, seq, err := mset.store.LoadPrevMsgMulti(sl, last, &smv) // loads/decompr
    if err == ErrStoreEOF || err != nil { break }
    last = seq - 1
    ...
    _, iName, sseq := streamAndSeq(bytesToString(ss))
    update(iName, sseq)                           // narrows sublist as sources
    if len(seqs) == expected { return }           // stop only when ALL sources
}
```

Cost: O(blocks back to the least-recently-active source) block loads → worst case the entire store.

After

```
mset.resetSourceInfo()
var smv StoreMsg
unresolved := map[string]*sourceInfo{}
```

```
// Phase 1: targeted index lookup per source.
for _, ssi := range mset.cfg.Sources {
    si := mset.sources[ssi.iname]
    if si == nil { continue }
    subj := storedSubjectForSource(ssi)           // FilterSubject, or transfor
    if subj == _EMPTY_ || subj == fwcs || subjectHasWildcard(subj) {
        unresolved[ssi.iname] = si                // can't attribute by a singl
        continue
    }
    sm, err := mset.store.LoadLastMsg(subj, &smv)  // O(1) via psim/fss, returns
    if err != nil || sm == nil || len(sm.hdr) == 0 {
        unresolved[ssi.iname] = si
        continue
    }
    if ss := sliceHeader(JSStreamSource, sm.hdr); len(ss) > 0 {
        if _, iName, sseq := streamAndSeq(bytesToString(ss)); iName == ssi.iname
            si.sseq, si.dseq = sseq, 0            // verified — resolved withou
            continue
        }
    }
    unresolved[ssi.iname] = si
}

// Phase 2: only the leftovers pay the (now tiny) reverse scan.
if len(unresolved) > 0 {
    mset.reverseScanForSources(unresolved)        // = today's loop, scoped to
}
```

Cost: Phase 1 = O(S) targeted lookups (recent sources share/cache the last block; a quiet source is **one** targeted load, not a walk). Phase 2 only runs for ambiguous sources, with a smaller sublist.

> Multi-destination transform sources: call `LoadLastMsg` for each destination and keep the highest origin `sseq`, or defer them to Phase 2. Either is fine; deferring is simplest to start.

6. Before → after (complexity)

	Before	After
Index used	none (generic message walk)	`psim` / `fss` per-subject last-block index
Blocks loaded	back to least-active source (≤ all blocks)	~1 per source, only the blocks actually holding a source's last msg
Sensitivity to a **quiet** source	drags the scan back across the whole store	none — jumps straight to its block
Sensitivity to **store size**	linear in blocks scanned	none
Worst case still possible?	normal case	only if every source is `>` /wildcard/shared-subject/pre-2.10 (→ Phase 2 == today)
Held lock	`mset.mu` for the whole walk	`mset.mu` for O(S) lookups; Phase 2 only for leftovers

The edge→hub topology (each edge mapped to a distinct subject/domain) lands entirely in Phase 1 → the backward scan disappears for that case.

7. Correctness & edge cases

- **Same semantics.** Today's scan records, per source, the **most recent** stored message's origin seq (first hit scanning backward). `LoadLastMsg` returns exactly that message. Deleted/interior messages are handled by `loadLast` (it skips `dmap` entries and walks to the previous block if needed), matching the scan's behaviour.
- **Shared subjects / direct publishes.** Resolved safely by the header-iname verification → fall to Phase 2.
- **Wildcard / empty (`>`) filters.** Can't be attributed to one source by subject → Phase 2 (same as today for them).
- **Pre-2.10 headers** (no `iname`, only stream name): verification won't match → Phase 2, which keeps the existing stream-name matching path.
- **memstore.** `MemStore` has the simpler linear `LoadPrevMsgMulti`; `LoadLastMsg` there is also index-light, but memstore streams are small, so Phase 2 cost is negligible. (We can add a memstore `loadLast` fast path later if needed.)
- **`setStartingSequenceForSources`** (the `STREAM.UPDATE` path, `stream.go:4566`) gets the same Phase 1 treatment for the subset of sources it processes.

8. Risks

- **Index freshness.** `psim` / `fss` may lazily need a recalculation (`lastNeedsUpdate` , `recalculateForSubj`); `loadLast` / `MultiLastSeqs` already handle that, so we inherit correct behaviour.
- **Behavioural parity.** Phase 1 must produce identical `si.sseq` to the old scan for non-ambiguous sources; this is the core test (below). Keeping Phase 2 byte-for-byte as today bounds the blast radius.
- **Encryption/compression.** Phase 1 still loads the one block holding a source's last message (decrypt/decompress), but only that block — no change in correctness, large reduction in volume.

9. Testing & validation

Prototype status — implemented ✅

A working prototype of the two-phase resolver is implemented in `startingSequenceForSources` (`server/stream.go`), with:

- `TestJetStreamStartingSequenceForSourcesIndexFastPath` (`server/jetstream_sourcing_resume_test.go`) — three distinct-subject sources (index fast path) plus one **subject-transform** source (phase 2 fallback), each with a different origin depth and buried under 20k direct publishes. Asserts every recovered `si.sseq` equals the expected last origin sequence. **Passes.**
- `BenchmarkJetStreamScanForSources` (existing, single source) and a new `BenchmarkJetStreamScanForSourcesMulti` (16 sources spread across the store).

Measured (filestore, single-server)

Benchmark	Before (reverse scan)	After (phase 1)	Speedup
`ScanForSources` — 1 source, ~100k buried msgs	~8.5 µs/op	~5.5 µs/op	~1.5×
`ScanForSourcesMulti` — 16 sources, spread	~110.8 µs/op	~14.0 µs/op	**~7.9×**

The single-source gap is modest because `LoadPrevMsgMulti` already skips non-matching messages per block; the win scales with the number of sources (each avoids a re-walk via a direct `psim` jump), which is exactly the edge→hub fan-in case.

Still to do

- **Ambiguity coverage:** add cases for sources sharing a subject, a `>` source, a direct-publish overlap, and a pre-2.10 header → assert correct fallback + sequences (the prototype handles these via the header check; tests should lock the behaviour in).
- **Apply to `setStartingSequenceForSources`** (the `STREAM.UPDATE` twin, `stream.go:4566`).
- **Transform fast path (optional):** resolve single-concrete-destination transform sources in phase 1 too (currently deferred to phase 2).
- Run the full sourcing suite with `-race`.

10. Relationship to the durable-consumer proposal

This change is **orthogonal and complementary** to the durable-consumer/`si.sseq`-persistence ideas in `sourcing-durable-resume.md`:

- Durable consumers / persisted `si.sseq` aim to **skip** resume work entirely (when state survives).
- This proposal makes the **fallback resume itself cheap** — which still runs on cold start, on snapshot restore, for limits/clustered streams, and any time persisted state is missing.

Recommended order: land this index-based resume first (self-contained, no protocol/state change, helps every retention type today), then layer the persistence/durable improvements on top.

Appendix — key references

Symbol	File:line	Role
`startingSequenceForSources`	`stream.go:4694`	the scan to replace (Phase 1 + Phase 2)
unconditional call	`stream.go:4821`	runs on every leader election / restart
`setStartingSequenceForSources`	`stream.go:4566`	update-path twin, same treatment
sublist build (stored subjects)	`stream.go:4742-4757`	defines a source's stored subject(s)
`LoadLastMsg` / `loadLast`	`filestore.go:9048` / `8939`	index-based last-msg-for-subject (returns header)

Symbol	File:line	Role
`MultiLastSeqs`	`filestore.go:3806`	batched last-seq-per-filter via index
`psi` / `fs.psim`	`filestore.go:168` / `195`	per-subject `{total, fblk, lblk}`; store-wide subject→blocks index
`mb.fss` / `SimpleState`	`filestore.go:239` / `store.go:180`	per-block subject→ `{Msgs, First, Last}` index
`psim` maintenance on write	`filestore.go:4933-4943`	how `total` / `lblk` are kept current
`LoadPrevMsgMulti`	`filestore.go:9403` / `memstore.go:1991`	the backward walk used by Phase 2
`Nats-Stream-Source` header	`stream.go:635`	constant; the per-message source provenance
`genSourceHeader` / `streamAndSeq`	`stream.go:4470` / `4545`	writes / parses origin stream, iname, origin seq